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Guangzhi Du, Yao Rong, Liyun Zuo



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Numerical analysis of two-grid decoupled finite element algorithm for a coupled Darcy/Darcy fracture model *

Guangzhi Du[†] Yao Rong[‡] Liyun Zuo[§]

Abstract

In this article, a two-grid decoupled finite element method with backtracking for a coupled Darcy/Darcy fracture model is proposed and studied. Error estimates for the pressure in both H^1 norm and L^2 norm are established. To arrive at error bound for the pressure in L^2 norm, a global coarse grid correction is carried out.

Key Words: Darcy/Darcy fracture model; Two-grid decoupled method; Finite element method; Backtracking technique

1 Introduction

Recently, the fluid flow in porous media with fractures has become one hot topic. Since the permeabilities of fractures are much higher or lower than that in the surrounding matrix, fractures are usually thought of privileged channels or barriers. It is difficult to develop corresponding mathematical theory of this coupled model, particularly when we consider interactions between surrounding domains and fractures. To simplify this coupled problem, one model [1] was developed for high permeability fracture problems by viewing the fracture as one less dimension comparing with the size of the whole domain. Whereafter, a modified model [5] was proposed for both large and small permeability of fractures. Up to now, lots of numerical schemes have been proposed and studied for the fracture models, such as the domain decomposition method [4], the two-grid decoupled methods [2, 6], the mixed nonconforming finite element method [3] and so on.

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[†]E-mail: gzdu@sdu.edu.cn. School of Mathematics and Statistics, Shandong Normal University, Jinan 250014, China.

[‡]E-mail: yr926@outlook.com. College of Engineering Physics, Shenzhen Technology University, Shenzhen, Guangdong 518118, China.

[§]E-mail: yeziliyun@126.com. School of Mathematical Sciences, University of Jinan, Jinan 250022, China.

The key idea of the two-grid decoupled method [2] is to solve the coupled problem on a relatively coarse grid, then decouple the mixed problem into three independent subproblems on a fine grid via the coarse approximation. Both the theoretical results and numerical results have validated that it is efficient and effective. However, the classical two-grid decoupled method only derived the error convergent in H^1 norm. In this paper, we aim to propose one modified two-grid algorithm to obtain the convergence order of the errors in L^2 norm. The crucial idea is to carry out one more coarse grid correction. Theoretical analysis shows that the coarse correction would help us to arrive at the optimal error bound in L^2 norm.

The rest of this paper is organized as follows. The mixed Darcy/Darcy fracture model is introduced in Section 2 and the modified two-grid decoupled method is proposed and analyzed in Section 3.

2 The mixed Darcy/Darcy fracture model

The coupled Darcy/Darcy fracture problem comprises of two 2-D linear elliptic problems in surrounding domains with a 1-D linear elliptic problem on the fracture. Let $\Omega = [a_1, a_2] \times [b_1, b_2]$ be a rectangular domain with the boundary $\Gamma = \partial\Omega$, and $\gamma = \{x = x_f\} \times [b_1, b_2]$ be the 1-D surface such that

$$\Omega_1 = [a_1, \gamma) \times [b_1, b_2], \quad \Omega_2 = (\gamma, a_2] \times [b_1, b_2], \quad \Omega \setminus \gamma = \Omega_1 \cup \Omega_2.$$

For $i = 1, 2$, deote $\Gamma_i = \partial\Omega_i \cap \Gamma$ and let $\vec{u}_i, p_i$ and u_f, p_f stand for the velocity and pressure on Ω_i and γ , respectively. In both the surrounding matrix and the fracture, the flow is governed by the Darcy's law.

$$\left\{ \begin{array}{ll} \vec{u}_i = -\mathbf{K}_i \nabla p_i & \text{in } \Omega_i, \quad i = 1, 2, \\ \operatorname{div} \vec{u}_i = q_i & \text{in } \Omega_i, \quad i = 1, 2, \\ u_f = -k_{f,y} d \frac{\partial p_f}{\partial y} & \text{in } \gamma, \\ \frac{\partial p_f}{\partial y} = q_f + (\vec{u}_1 \cdot \vec{n}_1|_\gamma + \vec{u}_2 \cdot \vec{n}_2|_\gamma) & \text{in } \gamma, \\ -\xi \vec{u}_i \cdot \vec{n}_i + \alpha_f p_i = \alpha_f p_f - (1 - \xi) \vec{u}_{i+1} \cdot \vec{n}_{i+1} & \text{in } \gamma, \quad i = 1, 2, \\ p_i = \bar{p}_i & \text{on } \Gamma_i, \quad i = 1, 2, \\ p_f = \bar{p}_f & \text{on } \partial\gamma, \end{array} \right. \quad (2.1)$$

where $\vec{n}_i$ represents the unit normal vector on the interface γ directed outward from Ω_i , and $\vec{n}_1 = -\vec{n}_2$. The index i varies in $\mathbb{Z}/2\mathbb{Z}$ such that $2+1=1$. d is the thickness of the fracture, $\alpha_f = \frac{2k_{f,x}}{d}$ and $\xi \in (\frac{1}{2}, 1]$. $\mathbf{K}_i, \mathbf{K}_f = \operatorname{diag}(k_{f,x}, k_{f,y})$ are symmetric matrices representing the permeability of flow in Ω_i and γ . Furthermore, there exist two positive constants $k_{max} > k_{min} > 0$ satisfying

$$\begin{aligned} k_{min} |\vec{\zeta}|^2 &\leq \vec{\zeta}^T \mathbf{K}_i \vec{\zeta} \leq k_{max} |\vec{\zeta}|^2, \quad \forall \vec{\zeta} \in \mathbb{R}^2, \\ k_{min} &\leq (k_{f,y} d), \quad (k_{f,x}/d) \leq k_{max}. \end{aligned}$$

In this paper, for simplicity, we shall impose the homogeneous boundary conditions on the pressure, namely,

$$p_i = 0 \text{ on } \Gamma_i, \quad p_f = 0 \text{ on } \partial\gamma.$$

The standard Sobolev spaces $W^{m,s}$ equipped with the norm $\|\cdot\|_{W^{m,s}}$ and the semi-norm $|\cdot|_{W^{m,s}}$ will be used. If $s = 2$, it is the classical Hilbert space H^m .

Denote the following spaces

$$\begin{aligned} W_i &= \{w_i \in H^1(\Omega_i) | w_i = 0, \quad \text{on } \Gamma_i\}, \quad i = 1, 2, \\ W_f &= \{w_f \in H^1(\gamma) | w_f = 0, \quad \text{on } \partial\gamma\}, \\ W &= \{w = (w_1, w_2, w_f) \in W_1 \times W_2 \times W_f\}, \end{aligned}$$

equipped with the norms

$$\begin{aligned} \|w\|_W &= \sum_{i=1,2} \|w_i\|_{H^1(\Omega_i)} + \|w_f\|_{H^1(\gamma)}, \\ \|w\|_0 &= \sum_{i=1,2} \|w_i\|_{L^2(\Omega_i)} + \|w_f\|_{L^2(\gamma)}. \end{aligned}$$

Then the weak formulation of the coupled fracture problem (2.1) is to seek $p = (p_1, p_2, p_f) \in W$ such that

$$a(p, w) = \sum_{i=1,2} (q_i, w_i)_{\Omega_i} + (q_f, w_f)_\gamma, \quad \forall w = (w_1, w_2, w_f) \in W, \quad (2.2)$$

where

$$\begin{aligned} a(p, w) &= \sum_{i=1,2} (\mathbf{K}_i \nabla p_i, \nabla w_i)_{\Omega_i} + (k_{f,y} d \frac{\partial p_f}{\partial y}, \frac{\partial w_f}{\partial y})_\gamma \\ &\quad + \sum_{i=1,2} (\frac{\alpha_f}{2\xi - 1} (\xi p_i + (1 - \xi) p_{i+1} - p_f), w_i)_\gamma - (\frac{\alpha_f}{2\xi - 1} (p_1 + p_2 - 2p_f), w_f)_\gamma. \end{aligned} \quad (2.3)$$

$(\cdot, \cdot)_D$ means the L^2 inner product on any given domain D .

It is clear that $a(\cdot, \cdot)$ is a bilinear form from $W \times W$ to $\mathbb{R}$, and from [2] we know that it is continuous and W -elliptic, thus the unique solution to the coupled problem could be derived. Furthermore, it follows

$$a(w, w) \geq k_{\min} (\sum_{i=1,2} |w_i|_{H^1(\Omega_i)}^2 + |w_f|_{H^1(\gamma)}^2) + 2k_{\min} \sum_{i=1,2} \|w_i - w_f\|_{L^2(\gamma)}^2. \quad (2.4)$$

And for some positive constant C , there holds

$$a(w, w) \geq C (\sum_{i=1,2} \|w_i\|_{H^1(\Omega_i)}^2 + \|w_f\|_{H^1(\gamma)}^2 + \sum_{i=1,2} \|w_i - w_f\|_{L^2(\gamma)}^2). \quad (2.5)$$

3 A two-grid decoupled algorithm with backtracking

Let us introduce the coupled scheme first.

Coupled scheme. Seek $p^h = (p_1^h, p_2^h, p_f^h) \in W^h$ such that

$$a(p^h, w^h) = \sum_{i=1,2} (q_i, w_i^h)_{\Omega_i} + (q_f, w_f^h)_{\gamma}, \quad \forall w^h = (w_1^h, w_2^h, w_f^h) \in W^h, \quad (3.1)$$

in which $W_i^h (i = 1, 2), W_f^h, W^h$ are the finite element spaces of W_i, W_f, W , respectively, and they could be constructed as described in [2]. For the coupled scheme, there holds

$$\|p - p^h\|_W \leq Ch, \quad (3.2)$$

$$\|p - p^h\|_0 \leq Ch^2, \quad (3.3)$$

in which $p_i \in H^2(\Omega_i)$ and $p_f \in H^2(\gamma)$ are true solutions of the coupled problem (2.1). The letter C is a positive constant independent of mesh size, and should be viewed as different values at its different appearances.

Two-grid decoupled algorithm with backtracking

- 1. Find a global coarse grid approximation $p^H = (p_1^H, p_2^H, p_f^H) \in W^H$ such that

$$a(p^H, w^H) = \sum_{i=1,2} (q_i, w_i^H)_{\Omega_i} + (q_f, w_f^H)_{\gamma}, \quad \forall w^H = (w_1^H, w_2^H, w_f^H) \in W^H.$$

- 2. Solve the following decoupled fine grid problem to get $\tilde{p}^h = (\tilde{p}_1^h, \tilde{p}_2^h, \tilde{p}_f^h) \in W^h$

$$\begin{aligned} & \sum_{i=1,2} (\mathbf{K}_i \nabla \tilde{p}_i^h, \nabla w_i^h)_{\Omega_i} + (k_{f,y} d \frac{\partial \tilde{p}_f^h}{\partial y}, \frac{\partial w_f^h}{\partial y})_{\gamma} \\ & + \sum_{i=1,2} (\frac{\alpha_f}{2\xi - 1} (\xi p_i^H + (1 - \xi) p_{i+1}^H - p_f^H), w_i^h)_{\gamma} - (\frac{\alpha_f}{2\xi - 1} (p_1^H + p_2^H - 2p_f^H), w_f^h)_{\gamma} \\ & = \sum_{i=1,2} (q_i, w_i^h)_{\Omega_i} + (q_f, w_f^h)_{\gamma}, \quad \forall w^h = (w_1^h, w_2^h, w_f^h) \in W^h. \end{aligned} \quad (3.4)$$

- 3. Carry out a coarse grid correction to get $e^H = (e_1^H, e_2^H, e_f^H) \in W^H$ such that

$$a(e^H, w^H) = \sum_{i=1,2} (q_i, w_i^H)_{\Omega_i} + (q_f, w_f^H)_{\gamma} - a(\tilde{p}^h, w^H), \quad \forall w^H = (w_1^H, w_2^H, w_f^H) \in W^H. \quad (3.5)$$

- 4. Obtain the final solution as

$$p^{H,h} = (p_1^{H,h}, p_2^{H,h}, p_f^{H,h}) = (\tilde{p}_1^h + e_1^H, \tilde{p}_2^h + e_2^H, \tilde{p}_f^h + e_f^H). \quad (3.6)$$

Remark. There are two mesh parameters in the above algorithm, and we always assume these two meshes align with each other, in this case, there holds $W^H \subset W^h$.

Theorem 3.1: *For the solution derived by the two-grid decoupled algorithm with backtracking, there holds*

$$\|p^h - p^{H,h}\|_W \leq CH^{\frac{3}{2}}, \quad (3.7)$$

$$\|p^h - p^{H,h}\|_0 \leq CH\|p^h - p^{H,h}\|_W. \quad (3.8)$$

Proof. Similar to the proof of Theorem 4.2 in [2], we could have

$$\|p^h - \tilde{p}^h\|_W \leq CH^{\frac{3}{2}}. \quad (3.9)$$

It is easy to verify that (3.5) could be rewritten as

$$\begin{aligned} a(e^H, w^H) &= \sum_{i=1,2} (q_i, w_i^H)_{\Omega_i} + (q_f, w_f^H)_{\gamma} - a(\tilde{p}^h, w^H) \\ &= a(p^h, w^H) - a(\tilde{p}^h, w^H) \\ &= a(p^h - \tilde{p}^h, w^H). \end{aligned}$$

Choosing $w^H = e^H$ in the above equality, using the fact $a(\cdot, \cdot)$ is continuous and W -elliptic, we have

$$\|e^H\|_W^2 \leq C\|p^h - \tilde{p}^h\|_W \|e^H\|_W,$$

which means

$$\|e^H\|_W \leq C\|p^h - \tilde{p}^h\|_W.$$

Therefore, there holds

$$\|p^h - p^{H,h}\|_W \leq \|p^h - \tilde{p}^h\|_W + \|e^H\|_W \leq C\|p^h - \tilde{p}^h\|_W \leq CH^{\frac{3}{2}}.$$

This is the first result (3.7).

To prove (3.8), we utilize the Aubin-Nitsche duality argument. For $p^h - p^{H,h} \in L^2(\Omega_1) \times L^2(\Omega_2) \times L^2(\gamma)$, there exists $\varphi = (\varphi_1, \varphi_2, \varphi_f) \in (H^2(\Omega_1) \times H^2(\Omega_2) \times H^2(\gamma)) \cap W$ such that

$$a(w, \varphi) = \sum_{i=1,2} (p_i^h - p_{i,H}^h, w_i)_{\Omega_i} + (p_f^h - p_{f,H}^h, w_f)_{\gamma}, \quad \forall w \in W. \quad (3.10)$$

For this problem, it is classical that

$$\sum_{i=1,2} \|\varphi_i\|_{H^2(\Omega_i)} + \|\varphi_f\|_{H^2(\gamma)} \leq C\|p^h - p_H^h\|_0.$$

In the following, we consider the H^1 -orthogonal projection $\Pi_H = (\Pi_{1,H}, \Pi_{2,H}, \Pi_{f,H}) : W \rightarrow W_H$ with respect to the coupled Darcy/Darcy fracture model. For given $w \in W$, there holds

$$a(v, w - \Pi_H w) = 0, \quad \forall v \in W,$$

and it follows that

$$\|w - \Pi_H w\|_W \leq CH \left(\sum_{i=1,2} \|w_i\|_{H^2(\Omega_i)} + \|w_f\|_{H^2(\gamma)} \right).$$

Substituting $w = p^h - p^{H,h}$ in (3.10) results in

$$\begin{aligned} & \sum_{i=1,2} \|p_i - p_{i,H}^h\|_{L^2(\Omega_i)}^2 + \|p_f - p_{f,H}^h\|_{L^2(\gamma)}^2 \\ &= a(p^h - p^{H,h}, \varphi) = a(p^h - p^{H,h}, \varphi - \Pi_H \varphi) \\ &\leq C \|p^h - p^{H,h}\|_W \|\varphi - \Pi_H \varphi\|_W \\ &\leq CH \|p^h - p^{H,h}\|_W \left(\sum_{i=1,2} \|\varphi_i\|_{H^2(\Omega_i)} + \|\varphi_f\|_{H^2(\gamma)} \right) \\ &\leq CH \|p^h - p^{H,h}\|_W \|p^h - p^{H,h}\|_0, \end{aligned}$$

which leads to (3.8). Hence we finish the proof. $\square$

By using the triangle inequality and (3.2)-(3.3), we could directly obtain the following results.

Theorem 3.2: *For the solution derived by the two-grid decoupled algorithm with backtracking, there holds*

$$\begin{aligned} \|p - p^{H,h}\|_W &\leq C(h + H^{\frac{3}{2}}), \\ \|p - p^{H,h}\|_0 &\leq C(h^2 + H\|p^h - p^{H,h}\|_W). \end{aligned}$$

Remark. We derive the estimates as $\|p^h - p^{H,h}\|_W \leq CH^{\frac{3}{2}}$ and $\|p - p^{H,h}\|_W \leq C(h + H^{\frac{3}{2}})$ since the proof of Theorem 4.2 in [2] shows that using the trace inequality $\|p_i^h - p_i^H\|_{L^2(\gamma)} \leq CH^{\frac{3}{2}}$ cuts down one in half order of the overall convergence order. It would be possible to improve these estimates if one could prove that $\|p_i^h - p_i^H\|_{L^2(\gamma)} \leq CH^2$. Similar estimate indeed holds for a single model Poisson problem on a convex domain with the Dirichlet boundary condition. However, it is not clear how to extend it to the present coupled Darcy/Darcy fracture model yet.

References

- [1] C. Alboin, J. Jaffre, J. Roberts, C. Serres, Domain decomposition for flow in fractured porous media, in: Domain Decomposition Methods in Sciences and Engineering, 1999, pp. 365–373.

- [2] S. Chen, H. Rui, A two-grid decoupled algorithm for fracture models, *Comput. Math. Appl.* 76(2018), 1161–1173.
- [3] N. Frih, V. Martin, J. Roberts, A. Saada, Modeling fractures as interfaces with nonmatching grids, *Comput. Geosci.* (2012), 1–18.
- [4] T. Hoang, C. Japhet, M. Kern, J. Roberts, Space-time domain decomposition for reduced fracture models in mixed formulation, *SIAM J. Numer. Anal.* 54(2016), 288–316.
- [5] V. Martin, J. Jaffre, J. Roberts, Modeling fractures and barriers as interfaces for flow in porous media, *SIAM J. Sci. Comput.* 26(2005), 1667–1691.
- [6] L. Zuo, G. Du, A novel two-grid method for a fracture model, *Appl. Math. Lett.* 125(2022), 107797.